

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z5,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z5 ZL: $\frac{Z_1 Z_5 Z_L g_m - Z_1 Z_L}{Z_1 Z_5 g_m + 2 Z_1 Z_L g_m + Z_1 + Z_5 + Z_L}$

$$H(z) = \frac{Z_1 Z_5 Z_L g_m - Z_1 Z_L}{Z_1 Z_5 g_m + 2 Z_1 Z_L g_m + Z_1 + Z_5 + Z_L}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_5 g_m - 1}{C_1 C_L R_5 s^2 + 2 g_m + s (C_1 + C_L R_5 g_m + C_L)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_L} \sqrt{R_5} \sqrt{g_m}}{C_1 + C_L R_5 g_m + C_L}$
wo: $\frac{\sqrt{2} \sqrt{g_m}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_5}}$
bandwidth: $\frac{C_1 + C_L R_5 g_m + C_L}{C_1 C_L R_5}$
K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.2 LP-2 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_5 R_L g_m - R_L}{C_1 C_L R_5 R_L s^2 + R_5 g_m + 2 R_L g_m + s (C_1 R_5 + C_1 R_L + C_L R_5 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_5} \sqrt{R_L} \sqrt{R_5 g_m + 2 R_L g_m + 1}}{C_1 R_5 + C_1 R_L + C_L R_5 R_L g_m + C_L R_L}$
wo: $\frac{\sqrt{R_5 g_m + 2 R_L g_m + 1}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_5} \sqrt{R_L}}$
bandwidth: $\frac{C_1 R_5 + C_1 R_L + C_L R_5 R_L g_m + C_L R_L}{C_1 C_L R_5 R_L}$
K-LP: $\frac{R_5 R_L g_m - R_L}{R_5 g_m + 2 R_L g_m + 1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.3 LP-3 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{C_1 C_L R_1 R_5 s^2 + 2R_1 g_m + s(C_1 R_1 + C_L R_1 R_5 g_m + C_L R_1 + C_L R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5} \sqrt{2R_1 g_m + 1}}{C_1 R_1 + C_L R_1 R_5 g_m + C_L R_1 + C_L R_5}$
 wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 + C_L R_1 R_5 g_m + C_L R_1 + C_L R_5}{C_1 C_L R_1 R_5}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.4 LP-4 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_5 R_L g_m - R_1 R_L}{C_1 C_L R_1 R_5 R_L s^2 + R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L + s(C_1 R_1 R_5 + C_1 R_1 R_L + C_L R_1 R_5 R_L g_m + C_L R_1 R_L + C_L R_5 R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5} \sqrt{R_L} \sqrt{R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L}}{C_1 R_1 R_5 + C_1 R_1 R_L + C_L R_1 R_5 R_L g_m + C_L R_1 R_L + C_L R_5 R_L}$
 wo: $\frac{\sqrt{R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 R_1 R_5 + C_1 R_1 R_L + C_L R_1 R_5 R_L g_m + C_L R_1 R_L + C_L R_5 R_L}{C_1 C_L R_1 R_5 R_L}$
 K-LP: $\frac{R_1 R_5 R_L g_m - R_1 R_L}{R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_1 R_L s + R_1 R_L g_m}{C_5 C_L R_1 R_L s^2 + R_1 g_m + s(2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L + C_L R_1 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{C_L} \sqrt{R_1} \sqrt{R_L} \sqrt{R_1 g_m + 1}}{2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L + C_L R_1 R_L g_m + C_L R_L}$
 wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_5} \sqrt{C_L} \sqrt{R_1} \sqrt{R_L}}$
 bandwidth: $\frac{2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L + C_L R_1 R_L g_m + C_L R_L}{C_5 C_L R_1 R_L}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_L}{2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L + C_L R_1 R_L g_m + C_L R_L}$
 Qz: None
 Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{C_5 C_L R_1 R_5 s^2 + 2R_1 g_m + s(2C_5 R_1 R_5 g_m + C_5 R_5 + C_L R_1 R_5 g_m + C_L R_1 + C_L R_5) + 1}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_5}\sqrt{C_L}\sqrt{R_1}\sqrt{R_5}\sqrt{2R_1g_m+1}}{2C_5R_1R_5g_m+C_5R_5+C_LR_1R_5g_m+C_LR_1+C_LR_5} \\
\text{wo: } & \frac{\sqrt{2R_1g_m+1}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_1}\sqrt{R_5}} \\
\text{bandwidth: } & \frac{2C_5R_1R_5g_m+C_5R_5+C_LR_1R_5g_m+C_LR_1+C_LR_5}{C_5C_LR_1R_5} \\
\text{K-LP: } & \frac{R_1R_5g_m-R_1}{2R_1g_m+1} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5R_1R_5}{2C_5R_1R_5g_m+C_5R_5+C_LR_1R_5g_m+C_LR_1+C_LR_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5R_5s+1}, \frac{R_L}{C_LR_Ls+1} \right)$

$$H(s) = \frac{-C_5R_1R_5R_Ls + R_1R_5R_Lg_m - R_1R_L}{C_5C_LR_1R_5R_Ls^2 + R_1R_5g_m + 2R_1R_Lg_m + R_1 + R_5 + R_L + s(2C_5R_1R_5R_Lg_m + C_5R_1R_5 + C_5R_5R_L + C_LR_1R_5R_Lg_m + C_LR_1R_L + C_LR_5R_L)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_5}\sqrt{C_L}\sqrt{R_1}\sqrt{R_5}\sqrt{R_L}\sqrt{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}{2C_5R_1R_5R_Lg_m+C_5R_1R_5+C_5R_5R_L+C_LR_1R_5R_Lg_m+C_LR_1R_L+C_LR_5R_L} \\
\text{wo: } & \frac{\sqrt{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_1}\sqrt{R_5}\sqrt{R_L}} \\
\text{bandwidth: } & \frac{2C_5R_1R_5R_Lg_m+C_5R_1R_5+C_5R_5R_L+C_LR_1R_5R_Lg_m+C_LR_1R_L+C_LR_5R_L}{C_5C_LR_1R_5R_L} \\
\text{K-LP: } & \frac{R_1R_5R_Lg_m-R_1R_L}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5R_1R_5R_L}{2C_5R_1R_5R_Lg_m+C_5R_1R_5+C_5R_5R_L+C_LR_1R_5R_Lg_m+C_LR_1R_L+C_LR_5R_L} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5s}, \frac{R_L}{C_LR_Ls+1} \right)$

$$H(s) = \frac{R_1R_Lg_m + s(C_5R_1R_5R_Lg_m - C_5R_1R_L)}{R_1g_m + s^2(C_5C_LR_1R_5R_Lg_m + C_5C_LR_1R_L + C_5C_LR_5R_L) + s(C_5R_1R_5g_m + 2C_5R_1R_Lg_m + C_5R_1 + C_5R_5 + C_5R_L + C_LR_1R_Lg_m + C_LR_L) + 1}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_5}\sqrt{C_L}R_1R_5\sqrt{R_L}g_m\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_5}\sqrt{C_L}R_1\sqrt{R_L}\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_5}\sqrt{C_L}R_5\sqrt{R_L}\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}}}{C_5R_1R_5g_m+2C_5R_1R_Lg_m+C_5R_1+C_5R_5+C_5R_L+C_LR_1R_Lg_m+C_LR_L} \\
\text{wo: } & \sqrt{\frac{R_1g_m+1}{C_5C_LR_1R_5R_Lg_m+C_5C_LR_1R_L+C_5C_LR_5R_L}} \\
\text{bandwidth: } & \frac{\sqrt{\frac{R_1g_m+1}{C_5C_LR_1R_5R_Lg_m+C_5C_LR_1R_L+C_5C_LR_5R_L}}(C_5R_1R_5g_m+2C_5R_1R_Lg_m+C_5R_1+C_5R_5+C_5R_L+C_LR_1R_Lg_m+C_LR_L)}{\sqrt{C_5}\sqrt{C_L}R_1R_5\sqrt{R_L}g_m\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_5}\sqrt{C_L}R_1\sqrt{R_L}\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}} + \sqrt{C_5}\sqrt{C_L}R_5\sqrt{R_L}\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5} + \frac{1}{R_1R_5g_m+R_1+R_5}}} \\
\text{K-LP: } & \frac{R_1R_Lg_m}{R_1g_m+1} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_5R_1R_5R_Lg_m-C_5R_1R_L}{C_5R_1R_5g_m+2C_5R_1R_Lg_m+C_5R_1+C_5R_5+C_5R_L+C_LR_1R_Lg_m+C_LR_L} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{1}{C_1s}, \infty, \infty, \infty, R_5, R_L + \frac{1}{C_Ls} \right)$

$$H(s) = \frac{R_5g_m + s(C_LR_5R_Lg_m - C_LR_L) - 1}{2g_m + s^2(C_1C_LR_5 + C_1C_LR_L) + s(C_1 + C_LR_5g_m + 2C_LR_Lg_m + C_L)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}\sqrt{g_m}\sqrt{R_5+R_L}}{C_1+C_LR_5g_m+2C_LR_Lg_m+C_L} \\
\text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1C_LR_5+C_1C_LR_L}} \\
\text{bandwidth: } & \frac{C_1+C_LR_5g_m+2C_LR_Lg_m+C_L}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_5+R_L}\sqrt{C_1C_LR_5+C_1C_LR_L}} \\
\text{K-LP: } & \frac{R_5g_m-1}{2g_m} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_LR_5R_Lg_m-C_LR_L}{C_1+C_LR_5g_m+2C_LR_Lg_m+C_L} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{-C_5 R_L s + R_L g_m}{C_1 C_5 R_L s^2 + g_m + s(C_1 + 2C_5 R_L g_m + C_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_L} \sqrt{g_m}}{C_1 + 2C_5 R_L g_m + C_5}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 + 2C_5 R_L g_m + C_5}{C_1 C_5 R_L}$
 K-LP: R_L
 K-HP: 0
 K-BP: $-\frac{C_5 R_L}{C_1 + 2C_5 R_L g_m + C_5}$
 Qz: None
 Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_L s + R_L g_m}{g_m + s^2(C_1 C_5 R_L + C_1 C_L R_L + C_5 C_L R_L) + s(C_1 + 2C_5 R_L g_m + C_5 + C_L R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{R_L} \sqrt{g_m} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L}}{C_1 + 2C_5 R_L g_m + C_5 + C_L R_L g_m}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1 C_5 R_L + C_1 C_L R_L + C_5 C_L R_L}}$
 bandwidth: $\frac{C_1 + 2C_5 R_L g_m + C_5 + C_L R_L g_m}{\sqrt{R_L} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{C_1 C_5 R_L + C_1 C_L R_L + C_5 C_L R_L}}$
 K-LP: R_L
 K-HP: 0
 K-BP: $-\frac{C_5 R_L}{C_1 + 2C_5 R_L g_m + C_5 + C_L R_L g_m}$
 Qz: None
 Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$

$$H(s) = \frac{-C_5 R_5 R_L s + R_5 R_L g_m - R_L}{C_1 C_5 R_5 R_L s^2 + R_5 g_m + 2R_L g_m + s(C_1 R_5 + C_1 R_L + 2C_5 R_5 R_L g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_5} \sqrt{R_L} \sqrt{R_5 g_m + 2R_L g_m + 1}}{C_1 R_5 + C_1 R_L + 2C_5 R_5 R_L g_m + C_5 R_5}$
 wo: $\frac{\sqrt{R_5 g_m + 2R_L g_m + 1}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_5} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 R_5 + C_1 R_L + 2C_5 R_5 R_L g_m + C_5 R_5}{C_1 C_5 R_5 R_L}$
 K-LP: $\frac{R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_5 R_L}{C_1 R_5 + C_1 R_L + 2C_5 R_5 R_L g_m + C_5 R_5}$
 Qz: None
 Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m - 1}{2g_m + s^2(C_1 C_5 R_5 + C_1 C_L R_5 + C_5 C_L R_5) + s(C_1 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{R_5} \sqrt{g_m} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L}}{C_1 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L}$

wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1 C_5 R_5 + C_1 C_L R_5 + C_5 C_L R_5}}$
bandwidth: $\frac{C_1 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L}{\sqrt{R_5} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{C_1 C_5 R_5 + C_1 C_L R_5 + C_5 C_L R_5}}$
K-LP: $\frac{R_5 g_m - 1}{2g_m}$
K-HP: 0
K-BP: $-\frac{C_5 R_5}{C_1 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_5 R_L s + R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + s^2 (C_1 C_5 R_5 R_L + C_1 C_L R_5 R_L + C_5 C_L R_5 R_L) + s (C_1 R_5 + C_1 R_L + 2C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_5} \sqrt{R_L} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{R_5 g_m + 2R_L g_m + 1}}{C_1 R_5 + C_1 R_L + 2C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L}$
wo: $\frac{\sqrt{R_5 g_m + 2R_L g_m + 1}}{\sqrt{C_1 C_5 R_5 R_L + C_1 C_L R_5 R_L + C_5 C_L R_5 R_L}}$
bandwidth: $\frac{C_1 R_5 + C_1 R_L + 2C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L}{\sqrt{R_5} \sqrt{R_L} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{C_1 C_5 R_5 R_L + C_1 C_L R_5 R_L + C_5 C_L R_5 R_L}}$
K-LP: $\frac{R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_5 R_L}{C_1 R_5 + C_1 R_L + 2C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L}$
Qz: None
Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{R_L g_m + s (C_5 R_5 R_L g_m - C_5 R_L)}{g_m + s^2 (C_1 C_5 R_5 + C_1 C_5 R_L) + s (C_1 + C_5 R_5 g_m + 2C_5 R_L g_m + C_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{g_m} \sqrt{R_5 + R_L}}{C_1 + C_5 R_5 g_m + 2C_5 R_L g_m + C_5}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_1 C_5 R_5 + C_1 C_5 R_L}}$
bandwidth: $\frac{C_1 + C_5 R_5 g_m + 2C_5 R_L g_m + C_5}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_5 + R_L} \sqrt{C_1 C_5 R_5 + C_1 C_5 R_L}}$
K-LP: R_L
K-HP: 0
K-BP: $\frac{C_5 R_5 R_L g_m - C_5 R_L}{C_1 + C_5 R_5 g_m + 2C_5 R_L g_m + C_5}$
Qz: None
Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s (C_L R_1 R_5 R_L g_m - C_L R_1 R_L)}{2R_1 g_m + s^2 (C_1 C_L R_1 R_5 + C_1 C_L R_1 R_L) + s (C_1 R_1 + C_L R_1 R_5 g_m + 2C_L R_1 R_L g_m + C_L R_1 + C_L R_5 + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5 + R_L} \sqrt{2R_1 g_m + 1}}{C_1 R_1 + C_L R_1 R_5 g_m + 2C_L R_1 R_L g_m + C_L R_1 + C_L R_5 + C_L R_L}$
wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_1 C_L R_1 R_5 + C_1 C_L R_1 R_L}}$
bandwidth: $\frac{C_1 R_1 + C_L R_1 R_5 g_m + 2C_L R_1 R_L g_m + C_L R_1 + C_L R_5 + C_L R_L}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5 + R_L} \sqrt{C_1 C_L R_1 R_5 + C_1 C_L R_1 R_L}}$
K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_L R_1 R_5 R_L g_m - C_L R_1 R_L}{C_1 R_1 + C_L R_1 R_5 g_m + 2C_L R_1 R_L g_m + C_L R_1 + C_L R_5 + C_L R_L}$
Qz: None
Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{-C_5 R_1 R_L s + R_1 R_L g_m}{C_1 C_5 R_1 R_L s^2 + R_1 g_m + s(C_1 R_1 + 2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_L} \sqrt{R_1 g_m + 1}}{C_1 R_1 + 2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L}$
 wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 R_1 + 2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L}{C_1 C_5 R_1 R_L}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_L}{C_1 R_1 + 2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L}$
 Qz: None
 Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_1 R_L s + R_1 R_L g_m}{R_1 g_m + s^2(C_1 C_5 R_1 R_L + C_1 C_L R_1 R_L + C_5 C_L R_1 R_L) + s(C_1 R_1 + 2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L + C_L R_1 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_L} \sqrt{R_1 g_m + 1} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L}}{C_1 R_1 + 2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L + C_L R_1 R_L g_m + C_L R_L}$
 wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1 C_5 R_1 R_L + C_1 C_L R_1 R_L + C_5 C_L R_1 R_L}}$
 bandwidth: $\frac{C_1 R_1 + 2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L + C_L R_1 R_L g_m + C_L R_L}{\sqrt{R_1} \sqrt{R_L} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{C_1 C_5 R_1 R_L + C_1 C_L R_1 R_L + C_5 C_L R_1 R_L}}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_L}{C_1 R_1 + 2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L + C_L R_1 R_L g_m + C_L R_L}$
 Qz: None
 Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$

$$H(s) = \frac{-C_5 R_1 R_5 R_L s + R_1 R_5 R_L g_m - R_1 R_L}{C_1 C_5 R_1 R_5 R_L s^2 + R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L + s(C_1 R_1 R_5 + C_1 R_1 R_L + 2C_5 R_1 R_5 R_L g_m + C_5 R_1 R_5 + C_5 R_5 R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5} \sqrt{R_L} \sqrt{R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L}}{C_1 R_1 R_5 + C_1 R_1 R_L + 2C_5 R_1 R_5 R_L g_m + C_5 R_1 R_5 + C_5 R_5 R_L}$
 wo: $\frac{\sqrt{R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 R_1 R_5 + C_1 R_1 R_L + 2C_5 R_1 R_5 R_L g_m + C_5 R_1 R_5 + C_5 R_5 R_L}{C_1 C_5 R_1 R_5 R_L}$
 K-LP: $\frac{R_1 R_5 R_L g_m - R_1 R_L}{R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_5 R_L}{C_1 R_1 R_5 + C_1 R_1 R_L + 2C_5 R_1 R_5 R_L g_m + C_5 R_1 R_5 + C_5 R_5 R_L}$
 Qz: None
 Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{2R_1 g_m + s^2(C_1 C_5 R_1 R_5 + C_1 C_L R_1 R_5 + C_5 C_L R_1 R_5) + s(C_1 R_1 + 2C_5 R_1 R_5 g_m + C_5 R_5 + C_L R_1 R_5 g_m + C_L R_1 + C_L R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_5} \sqrt{2R_1 g_m + 1} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L}}{C_1 R_1 + 2C_5 R_1 R_5 g_m + C_5 R_5 + C_L R_1 R_5 g_m + C_L R_1 + C_L R_5}$
 wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_1 C_5 R_1 R_5 + C_1 C_L R_1 R_5 + C_5 C_L R_1 R_5}}$
 bandwidth: $\frac{C_1 R_1 + 2C_5 R_1 R_5 g_m + C_5 R_5 + C_L R_1 R_5 g_m + C_L R_1 + C_L R_5}{\sqrt{R_1} \sqrt{R_5} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{C_1 C_5 R_1 R_5 + C_1 C_L R_1 R_5 + C_5 C_L R_1 R_5}}$

K-LP: $\frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_5}{C_1 R_1 + 2 C_5 R_1 R_5 g_m + C_5 R_5 + C_L R_1 R_5 g_m + C_L R_1 + C_L R_5}$
Qz: None
Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_1 R_5 R_L s + R_1 R_5 R_L g_m - R_1 R_L}{R_1 R_5 g_m + 2 R_1 R_L g_m + R_1 + R_5 + R_L + s^2 (C_1 C_5 R_1 R_5 R_L + C_1 C_L R_1 R_5 R_L + C_5 C_L R_1 R_5 R_L) + s (C_1 R_1 R_5 + C_1 R_1 R_L + 2 C_5 R_1 R_5 R_L g_m + C_5 R_1 R_5 + C_5 R_5 R_L + C_L R_1 R_5 R_L g_m + C_L R_1 R_L + C_L R_5 R_L)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_5} \sqrt{R_L} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{R_1 R_5 g_m + 2 R_1 R_L g_m + R_1 + R_5 + R_L}}{C_1 R_1 R_5 + C_1 R_1 R_L + 2 C_5 R_1 R_5 R_L g_m + C_5 R_1 R_5 + C_5 R_5 R_L + C_L R_1 R_5 R_L g_m + C_L R_1 R_L + C_L R_5 R_L}$
wo: $\frac{\sqrt{R_1 R_5 g_m + 2 R_1 R_L g_m + R_1 + R_5 + R_L}}{\sqrt{C_1 C_5 R_1 R_5 R_L + C_1 C_L R_1 R_5 R_L + C_5 C_L R_1 R_5 R_L}}$
bandwidth: $\frac{C_1 R_1 R_5 + C_1 R_1 R_L + 2 C_5 R_1 R_5 R_L g_m + C_5 R_1 R_5 + C_5 R_5 R_L + C_L R_1 R_5 R_L g_m + C_L R_1 R_L + C_L R_5 R_L}{\sqrt{R_1} \sqrt{R_5} \sqrt{R_L} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{C_1 C_5 R_1 R_5 R_L + C_1 C_L R_1 R_5 R_L + C_5 C_L R_1 R_5 R_L}}$
K-LP: $\frac{R_1 R_5 R_L g_m - R_1 R_L}{R_1 R_5 g_m + 2 R_1 R_L g_m + R_1 + R_5 + R_L}$
K-HP: 0
K-BP: $-\frac{C_5 R_1 R_5 R_L}{C_1 R_1 R_5 + C_1 R_1 R_L + 2 C_5 R_1 R_5 R_L g_m + C_5 R_1 R_5 + C_5 R_5 R_L + C_L R_1 R_5 R_L g_m + C_L R_1 R_L + C_L R_5 R_L}$
Qz: None
Wz: None

8.18 X-INVALID-NUMER-18 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{R_1 R_L g_m + s (C_5 R_1 R_5 R_L g_m - C_5 R_1 R_L)}{R_1 g_m + s^2 (C_1 C_5 R_1 R_5 + C_1 C_5 R_1 R_L) + s (C_1 R_1 + C_5 R_1 R_5 g_m + 2 C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_5 + C_5 R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5 + R_L} \sqrt{R_1 g_m + 1}}{C_1 R_1 + C_5 R_1 R_5 g_m + 2 C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_5 + C_5 R_L}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1 C_5 R_1 R_5 + C_1 C_5 R_1 R_L}}$
bandwidth: $\frac{C_1 R_1 + C_5 R_1 R_5 g_m + 2 C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_5 + C_5 R_L}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5 + R_L} \sqrt{C_1 C_5 R_1 R_5 + C_1 C_5 R_1 R_L}}$
K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: $\frac{C_5 R_1 R_5 R_L g_m - C_5 R_1 R_L}{C_1 R_1 + C_5 R_1 R_5 g_m + 2 C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_5 + C_5 R_L}$
Qz: None
Wz: None

8.19 X-INVALID-NUMER-19 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1) - 1}{2 g_m + s^2 (C_1 C_L R_1 R_5 g_m + C_1 C_L R_1 + C_1 C_L R_5) + s (2 C_1 R_1 g_m + C_1 + C_L R_5 g_m + C_L)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_L} R_1 R_5 g_m \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_1} \sqrt{C_L} R_1 \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_1} \sqrt{C_L} R_5 \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}}}{2 C_1 R_1 g_m + C_1 + C_L R_5 g_m + C_L}$
wo: $\sqrt{2} \sqrt{\frac{g_m}{C_1 C_L R_1 R_5 g_m + C_1 C_L R_1 + C_1 C_L R_5}}$
bandwidth: $\frac{\sqrt{2} \sqrt{\frac{g_m}{C_1 C_L R_1 R_5 g_m + C_1 C_L R_1 + C_1 C_L R_5}} (2 C_1 R_1 g_m + C_1 + C_L R_5 g_m + C_L)}{\sqrt{2} \sqrt{C_1} \sqrt{C_L} R_1 R_5 g_m \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_1} \sqrt{C_L} R_1 \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{2} \sqrt{C_1} \sqrt{C_L} R_5 \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}}}$
K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_5 g_m - C_1 R_1}{2 C_1 R_1 g_m + C_1 + C_L R_5 g_m + C_L}$
Qz: None
Wz: None

8.20 X-INVALID-NUMER-20 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_5 R_L g_m - R_L + s(C_1 R_1 R_5 R_L g_m - C_1 R_1 R_L)}{R_5 g_m + 2R_L g_m + s^2(C_1 C_L R_1 R_5 R_L g_m + C_1 C_L R_1 R_L + C_1 C_L R_5 R_L) + s(C_1 R_1 R_5 g_m + 2C_1 R_1 R_L g_m + C_1 R_1 + C_1 R_5 + C_1 R_L + C_L R_5 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} R_1 R_5 \sqrt{R_L g_m} \sqrt{\frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_L g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_1} \sqrt{C_L} R_1 \sqrt{R_L} \sqrt{\frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_L g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_1} \sqrt{C_L} R_5 \sqrt{R_L} \sqrt{\frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_L g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}}}{C_1 R_1 R_5 g_m + 2C_1 R_1 R_L g_m + C_1 R_1 + C_1 R_5 + C_1 R_L + C_L R_5 R_L g_m + C_L R_L}$

wo: $\sqrt{\frac{R_5 g_m + 2R_L g_m + 1}{C_1 C_L R_1 R_5 R_L g_m + C_1 C_L R_1 R_L + C_1 C_L R_5 R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_5 g_m + 2R_L g_m + 1}{C_1 C_L R_1 R_5 R_L g_m + C_1 C_L R_1 R_L + C_1 C_L R_5 R_L}} (C_1 R_1 R_5 g_m + 2C_1 R_1 R_L g_m + C_1 R_1 + C_1 R_5 + C_1 R_L + C_L R_5 R_L g_m + C_L R_L)}{\sqrt{C_1} \sqrt{C_L} R_1 R_5 \sqrt{R_L g_m} \sqrt{\frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_L g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_1} \sqrt{C_L} R_1 \sqrt{R_L} \sqrt{\frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_L g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + \sqrt{C_1} \sqrt{C_L} R_5 \sqrt{R_L} \sqrt{\frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{2R_L g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}}}$

K-LP: $\frac{R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + 1}$

K-HP: 0

K-BP: $\frac{C_1 R_1 R_5 R_L g_m - C_1 R_1 R_L}{C_1 R_1 R_5 g_m + 2C_1 R_1 R_L g_m + C_1 R_1 + C_1 R_5 + C_1 R_L + C_L R_5 R_L g_m + C_L R_L}$

Qz: None

Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, \infty, \infty, R_5, R_L)$

$$H(s) = \frac{R_1 R_5 R_L g_m - R_1 R_L}{R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{2R_1 g_m + s(C_L R_1 R_5 g_m + C_L R_1 + C_L R_5) + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_5 R_L g_m - R_1 R_L}{R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L + s(C_L R_1 R_5 R_L g_m + C_L R_1 R_L + C_L R_5 R_L)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s(C_L R_1 R_5 R_L g_m - C_L R_1 R_L)}{2R_1 g_m + s(C_L R_1 R_5 g_m + 2C_L R_1 R_L g_m + C_L R_1 + C_L R_5 + C_L R_L) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{-C_5 R_1 R_L s + R_1 R_L g_m}{R_1 g_m + s(2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_L) + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{C_5 C_L R_1 s^2 + s(2C_5 R_1 g_m + C_5 + C_L R_1 g_m + C_L)} + 1$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_1 R_L s^2 + R_1 g_m + s(-C_5 R_1 + C_L R_1 R_L g_m)}{s^2 (2C_5 C_L R_1 R_L g_m + C_5 C_L R_1 + C_5 C_L R_L) + s(2C_5 R_1 g_m + C_5 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$$

$$H(s) = \frac{-C_5 R_1 R_5 R_L s + R_1 R_5 R_L g_m - R_1 R_L}{R_1 R_5 g_m + 2R_1 R_L g_m + R_1 + R_5 + R_L + s(2C_5 R_1 R_5 R_L g_m + C_5 R_1 R_5 + C_5 R_5 R_L)}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L \right)$$

$$H(s) = \frac{R_1 R_L g_m + s(C_5 R_1 R_5 R_L g_m - C_5 R_1 R_L)}{R_1 g_m + s(C_5 R_1 R_5 g_m + 2C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_5 + C_5 R_L) + 1}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m + s(C_5 R_1 R_5 g_m - C_5 R_1)}{s^2 (C_5 C_L R_1 R_5 g_m + C_5 C_L R_1 + C_5 C_L R_5) + s(2C_5 R_1 g_m + C_5 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_5 C_L R_1 R_5 R_L g_m - C_5 C_L R_1 R_L) + s(C_5 R_1 R_5 g_m - C_5 R_1 + C_L R_1 R_L g_m)}{s^2 (C_5 C_L R_1 R_5 g_m + 2C_5 C_L R_1 R_L g_m + C_5 C_L R_1 + C_5 C_L R_5 + C_5 C_L R_L) + s(2C_5 R_1 g_m + C_5 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5, R_L \right)$$

$$H(s) = \frac{R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + s(C_1 R_5 + C_1 R_L) + 1}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 s + g_m}{s^2 (C_1 C_5 + C_1 C_L + C_5 C_L) + s(2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_L s^2 + g_m + s(-C_5 + C_L R_L g_m)}{C_1 C_5 C_L R_L s^3 + s^2 (C_1 C_5 + C_1 C_L + 2C_5 C_L R_L g_m + C_5 C_L) + s(2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_5 R_L s^2 + R_5 g_m + s(-C_5 R_5 + C_L R_5 R_L g_m - C_L R_L) - 1}{C_1 C_5 C_L R_5 R_L s^3 + 2g_m + s^2 (C_1 C_5 R_5 + C_1 C_L R_5 + C_1 C_L R_L + 2C_5 C_L R_5 R_L g_m + C_5 C_L R_5) + s(C_1 + 2C_5 R_5 g_m + C_L R_5 g_m + 2C_L R_L g_m + C_L)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m + s(C_5 R_5 g_m - C_5)}{C_1 C_5 C_L R_5 s^3 + s^2 (C_1 C_5 + C_1 C_L + C_5 C_L R_5 g_m + C_5 C_L) + s(2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L g_m + s (C_5 R_5 R_L g_m - C_5 R_L)}{C_1 C_5 C_L R_5 R_L s^3 + g_m + s^2 (C_1 C_5 R_5 + C_1 C_5 R_L + C_1 C_L R_L + C_5 C_L R_5 R_L g_m + C_5 C_L R_L) + s (C_1 + C_5 R_5 g_m + 2 C_5 R_L g_m + C_5 + C_L R_L g_m)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m + s^2 (C_5 C_L R_5 R_L g_m - C_5 C_L R_L) + s (C_5 R_5 g_m - C_5 + C_L R_L g_m)}{s^3 (C_1 C_5 C_L R_5 + C_1 C_5 C_L R_L) + s^2 (C_1 C_5 + C_1 C_L + C_5 C_L R_5 g_m + 2 C_5 C_L R_L g_m + C_5 C_L) + s (2 C_5 g_m + C_L g_m)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5, R_L \right)$$

$$H(s) = \frac{R_1 R_5 R_L g_m - R_1 R_L}{R_1 R_5 g_m + 2 R_1 R_L g_m + R_1 + R_5 + R_L + s (C_1 R_1 R_5 + C_1 R_1 R_L)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{s^2 (C_1 C_5 R_1 + C_1 C_L R_1 + C_5 C_L R_1) + s (2 C_5 R_1 g_m + C_5 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_1 R_L s^2 + R_1 g_m + s (-C_5 R_1 + C_L R_1 R_L g_m)}{C_1 C_5 C_L R_1 R_L s^3 + s^2 (C_1 C_5 R_1 + C_1 C_L R_1 + 2 C_5 C_L R_1 R_L g_m + C_5 C_L R_1 + C_5 C_L R_L) + s (2 C_5 R_1 g_m + C_5 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_1 R_5 R_L s^2 + R_1 R_5 g_m - R_1 + s (-C_5 R_1 R_5 + C_L R_1 R_5 R_L g_m - C_L R_1 R_L)}{C_1 C_5 C_L R_1 R_5 R_L s^3 + 2 R_1 g_m + s^2 (C_1 C_5 R_1 R_5 + C_1 C_L R_1 R_5 + C_1 C_L R_1 R_L + 2 C_5 C_L R_1 R_5 R_L g_m + C_5 C_L R_1 R_5 + C_5 C_L R_5 R_L) + s (C_1 R_1 + 2 C_5 R_1 R_5 g_m + C_5 R_5 + C_L R_1 R_5 g_m + 2 C_L R_1 R_L g_m + C_L R_1 + C_L R_5 + C_L R_L) + 1}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m + s (C_5 R_1 R_5 g_m - C_5 R_1)}{C_1 C_5 C_L R_1 R_5 s^3 + s^2 (C_1 C_5 R_1 + C_1 C_L R_1 + C_5 C_L R_1 R_5 g_m + C_5 C_L R_1 + C_5 C_L R_5) + s (2 C_5 R_1 g_m + C_5 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_1 R_L g_m + s (C_5 R_1 R_5 R_L g_m - C_5 R_1 R_L)}{C_1 C_5 C_L R_1 R_5 R_L s^3 + R_1 g_m + s^2 (C_1 C_5 R_1 R_5 + C_1 C_5 R_1 R_L + C_1 C_L R_1 R_L + C_5 C_L R_1 R_5 R_L g_m + C_5 C_L R_1 R_L + C_5 C_L R_5 R_L) + s (C_1 R_1 + C_5 R_1 R_5 g_m + 2 C_5 R_1 R_L g_m + C_5 R_1 + C_5 R_5 + C_5 R_L + C_L R_1 R_L g_m + C_L R_L) + 1}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_5 C_L R_1 R_5 R_L g_m - C_5 C_L R_1 R_L) + s (C_5 R_1 R_5 g_m - C_5 R_1 + C_L R_1 R_L g_m)}{s^3 (C_1 C_5 C_L R_1 R_5 + C_1 C_5 C_L R_1 R_L) + s^2 (C_1 C_5 R_1 + C_1 C_L R_1 + C_5 C_L R_1 R_5 g_m + 2 C_5 C_L R_1 R_L g_m + C_5 C_L R_1 + C_5 C_L R_5 + C_5 C_L R_L) + s (2 C_5 R_1 g_m + C_5 + C_L R_1 g_m + C_L)}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, R_L \right)$$

$$H(s) = \frac{R_5 R_L g_m - R_L + s (C_1 R_1 R_5 R_L g_m - C_1 R_1 R_L)}{R_5 g_m + 2 R_L g_m + s (C_1 R_1 R_5 g_m + 2 C_1 R_1 R_L g_m + C_1 R_1 + C_1 R_5 + C_1 R_L) + 1}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_1 C_5 R_1 s^2 + g_m + s (C_1 R_1 g_m - C_5)}{C_1 C_5 C_L R_1 s^3 + s^2 (2C_1 C_5 R_1 g_m + C_1 C_5 + C_1 C_L R_1 g_m + C_1 C_L + C_5 C_L) + s (2C_5 g_m + C_L g_m)}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_L s^2 + R_L g_m + s (C_1 R_1 R_L g_m - C_5 R_L)}{C_1 C_5 C_L R_1 R_L s^3 + g_m + s^2 (2C_1 C_5 R_1 R_L g_m + C_1 C_5 R_1 + C_1 C_5 R_L + C_1 C_L R_1 R_L g_m + C_1 C_L R_L + C_5 C_L R_L) + s (C_1 R_1 g_m + C_1 + 2C_5 R_L g_m + C_5 + C_L R_L g_m)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_1 C_5 C_L R_1 R_L s^3 + g_m + s^2 (-C_1 C_5 R_1 + C_1 C_L R_1 R_L g_m - C_5 C_L R_L) + s (C_1 R_1 g_m - C_5 + C_L R_L g_m)}{s^3 (2C_1 C_5 C_L R_1 R_L g_m + C_1 C_5 C_L R_1 + C_1 C_5 C_L R_L) + s^2 (2C_1 C_5 R_1 g_m + C_1 C_5 + C_1 C_L R_1 g_m + C_1 C_L + 2C_5 C_L R_L g_m + C_5 C_L) + s (2C_5 g_m + C_L g_m)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 s^2 + R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5) - 1}{C_1 C_5 C_L R_1 R_5 s^3 + 2g_m + s^2 (2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + C_1 C_L R_1 R_5 g_m + C_1 C_L R_1 + C_1 C_L R_5 + C_5 C_L R_5) + s (2C_1 R_1 g_m + C_1 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 R_L s^2 + R_5 R_L g_m - R_L + s (C_1 R_1 R_5 R_L g_m - C_1 R_1 R_L - C_5 R_5 R_L)}{C_1 C_5 C_L R_1 R_5 R_L s^3 + R_5 g_m + 2R_L g_m + s^2 (2C_1 C_5 R_1 R_5 R_L g_m + C_1 C_5 R_1 R_5 + C_1 C_5 R_5 R_L + C_1 C_L R_1 R_5 R_L g_m + C_1 C_L R_1 R_L + C_1 C_L R_5 R_L + C_5 C_L R_5 R_L) + s (C_1 R_1 R_5 g_m + 2C_1 R_1 R_L g_m + C_1 R_1 + C_1 R_5 + C_1 R_L + 2C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L)}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_1 C_5 C_L R_1 R_5 R_L s^3 + R_5 g_m + s^2 (-C_1 C_5 R_1 R_5 + C_1 C_L R_1 R_5 R_L g_m - C_1 C_L R_1 R_L - C_5 C_L R_5 R_L) + s (C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5 + C_L R_5 R_L g_m - C_L R_L) - 1}{2g_m + s^3 (2C_1 C_5 C_L R_1 R_5 R_L g_m + C_1 C_5 C_L R_1 R_5 + C_1 C_5 C_L R_5 R_L) + s^2 (2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + C_1 C_L R_1 R_5 g_m + 2C_1 C_L R_1 R_L g_m + C_1 C_L R_1 + C_1 C_L R_5 + C_1 C_L R_L + 2C_5 C_L R_5 R_L g_m + C_5 C_L R_5) + s (2C_1 R_1 g_m + C_1 + 2C_5 R_5 g_m + C_L R_5 g_m + 2C_L R_L g_m + C_L)}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{s^3 (C_1 C_5 C_L R_1 R_5 g_m + C_1 C_5 C_L R_1 + C_1 C_5 C_L R_5) + s^2 (2C_1 C_5 R_1 g_m + C_1 C_5 + C_1 C_L R_1 g_m + C_1 C_L + C_5 C_L R_5 g_m + C_5 C_L) + s (2C_5 g_m + C_L g_m)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m + s^2 (C_1 C_5 R_1 R_5 R_L g_m - C_1 C_5 R_1 R_L) + s (C_1 R_1 R_L g_m + C_5 R_5 R_L g_m - C_5 R_L)}{g_m + s^3 (C_1 C_5 C_L R_1 R_5 R_L g_m + C_1 C_5 C_L R_1 R_L + C_1 C_5 C_L R_5 R_L) + s^2 (C_1 C_5 R_1 R_5 g_m + 2C_1 C_5 R_1 R_L g_m + C_1 C_5 R_1 + C_1 C_5 R_5 + C_1 C_5 R_L + C_1 C_L R_1 R_L g_m + C_1 C_L R_L + C_5 C_L R_5 R_L g_m + C_5 C_L R_L) + s (C_1 R_1 g_m + C_1 + C_5 R_5 g_m + 2C_5 R_L g_m + C_5 + C_L R_L g_m)}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m + s^3 (C_1 C_5 C_L R_1 R_5 R_L g_m - C_1 C_5 C_L R_1 R_L) + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_1 C_L R_1 R_L g_m + C_5 C_L R_5 R_L g_m - C_5 C_L R_L) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5 + C_L R_L g_m)}{s^3 (C_1 C_5 C_L R_1 R_5 g_m + 2C_1 C_5 C_L R_1 R_L g_m + C_1 C_5 C_L R_1 + C_1 C_5 C_L R_5 + C_1 C_5 C_L R_L) + s^2 (2C_1 C_5 R_1 g_m + C_1 C_5 + C_1 C_L R_1 g_m + C_1 C_L + C_5 C_L R_5 g_m + 2C_5 C_L R_L g_m + C_5 C_L) + s (2C_5 g_m + C_L g_m)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 C_L R_1 R_5 R_L s^2 + R_1 R_5 g_m - R_1 + s(-C_5 R_1 R_5 + C_L R_1 R_5 R_L g_m - C_L R_1 R_L)}{2 R_1 g_m + s^2 (2 C_5 C_L R_1 R_5 R_L g_m + C_5 C_L R_1 R_5 + C_5 C_L R_5 R_L) + s (2 C_5 R_1 R_5 g_m + C_5 R_5 + C_L R_1 R_5 g_m + 2 C_L R_1 R_L g_m + C_L R_1 + C_L R_5 + C_L R_L) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_5}\sqrt{C_L}R_1\sqrt{R_5}R_Lg_m\sqrt{\frac{2R_1g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_5}\sqrt{C_L}R_1\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}R_L\sqrt{\frac{2R_1g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}}{2C_5R_1R_5g_m+C_5R_5+C_LR_1R_5g_m+2C_LR_1R_Lg_m+C_LR_1+C_LR_5+C_LR_L} \\ \text{wo: } & \sqrt{\frac{2R_1g_m+1}{2C_5C_LR_1R_5R_Lg_m+C_5C_LR_1R_5+C_5C_LR_5R_L}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{2R_1g_m+1}{2C_5C_LR_1R_5R_Lg_m+C_5C_LR_1R_5+C_5C_LR_5R_L}}(2C_5R_1R_5g_m+C_5R_5+C_LR_1R_5g_m+2C_LR_1R_Lg_m+C_LR_1+C_LR_5+C_LR_L)}{2\sqrt{C_5}\sqrt{C_L}R_1\sqrt{R_5}R_Lg_m\sqrt{\frac{2R_1g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_5}\sqrt{C_L}R_1\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}R_L\sqrt{\frac{2R_1g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}} \\ \text{K-LP: } & \frac{R_1R_5g_m-R_1}{2R_1g_m+1} \\ \text{K-HP: } & -\frac{R_1R_L}{2R_1R_Lg_m+R_1+R_L} \\ \text{K-BP: } & \frac{-C_5R_1R_5+C_LR_1R_5R_Lg_m-C_LR_1R_L}{2C_5R_1R_5g_m+C_5R_5+C_LR_1R_5g_m+2C_LR_1R_Lg_m+C_LR_1+C_LR_5+C_LR_L} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-R_5g_m+1}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}\sqrt{R_L}} \end{aligned}$$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 C_L R_1 R_5 R_L g_m - C_1 C_L R_1 R_L) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_L R_5 R_L g_m - C_L R_L) - 1}{2 g_m + s^2 (C_1 C_L R_1 R_5 g_m + 2 C_1 C_L R_1 R_L g_m + C_1 C_L R_1 + C_1 C_L R_5 + C_1 C_L R_L) + s (2 C_1 R_1 g_m + C_1 + C_L R_5 g_m + 2 C_L R_L g_m + C_L)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_5g_m\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_Lg_m\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_5\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_L\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}}{2C_1R_1g_m+C_1+C_LR_5g_m+2C_LR_Lg_m+C_L} \\ \text{wo: } & \sqrt{2}\sqrt{\frac{g_m}{C_1C_LR_1R_5g_m+2C_1C_LR_1R_Lg_m+C_1C_LR_1+C_1C_LR_5+C_1C_LR_L}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{\frac{g_m}{C_1C_LR_1R_5g_m+2C_1C_LR_1R_Lg_m+C_1C_LR_1+C_1C_LR_5+C_1C_LR_L}}(2C_1R_1g_m+C_1+C_LR_5g_m+2C_LR_Lg_m+C_L)}{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_5g_m\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_Lg_m\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_5\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_L\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}} \\ \text{K-LP: } & \frac{R_5g_m-1}{2g_m} \\ \text{K-HP: } & \frac{R_1R_5R_Lg_m-R_1R_L}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L} \\ \text{K-BP: } & \frac{C_1R_1R_5g_m-C_1R_1+C_LR_5R_Lg_m-C_LR_L}{2C_1R_1g_m+C_1+C_LR_5g_m+2C_LR_Lg_m+C_L} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{1}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_L}} \end{aligned}$$

10.3 X-INVALID-WZ-3 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_L s^2 + R_L g_m + s (C_1 R_1 R_L g_m - C_5 R_L)}{g_m + s^2 (2 C_1 C_5 R_1 R_L g_m + C_1 C_5 R_1 + C_1 C_5 R_L) + s (C_1 R_1 g_m + C_1 + 2 C_5 R_L g_m + C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_1}\sqrt{C_5}R_1R_Lg_m\sqrt{\frac{g_m}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_1}\sqrt{C_5}R_L\sqrt{\frac{g_m}{2R_1R_Lg_m+R_1+R_L}}}{C_1R_1g_m+C_1+2C_5R_Lg_m+C_5} \\ \text{wo: } & \sqrt{\frac{g_m}{2C_1C_5R_1R_Lg_m+C_1C_5R_1+C_1C_5R_L}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{g_m}{2C_1C_5R_1R_Lg_m+C_1C_5R_1+C_1C_5R_L}}(C_1R_1g_m+C_1+2C_5R_Lg_m+C_5)}{2\sqrt{C_1}\sqrt{C_5}R_1R_Lg_m\sqrt{\frac{g_m}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_1}\sqrt{C_5}R_L\sqrt{\frac{g_m}{2R_1R_Lg_m+R_1+R_L}}} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & -\frac{R_1R_L}{2R_1R_Lg_m+R_1+R_L} \\ \text{K-BP: } & \frac{C_1R_1R_Lg_m-C_5R_L}{C_1R_1g_m+C_1+2C_5R_Lg_m+C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-g_m}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}} \end{aligned}$$

10.4 X-INVALID-WZ-4 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 R_L s^2 + R_5 R_L g_m - R_L + s (C_1 R_1 R_5 R_L g_m - C_1 R_1 R_L - C_5 R_5 R_L)}{R_5 g_m + 2 R_L g_m + s^2 (2 C_1 C_5 R_1 R_5 R_L g_m + C_1 C_5 R_1 R_5 + C_1 C_5 R_5 R_L) + s (C_1 R_1 R_5 g_m + 2 C_1 R_1 R_L g_m + C_1 R_1 + C_1 R_5 + C_1 R_L + 2 C_5 R_5 R_L g_m + C_5 R_5) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}R_Lg_m\sqrt{\frac{R_5g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{2R_Lg_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{R_5g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{2R_Lg_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_1}\sqrt{C_5}\sqrt{R_5}R_L\sqrt{\frac{R_5g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{2R_Lg_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}}{C_1R_1R_5g_m+2C_1R_1R_Lg_m+C_1R_1+C_1R_5+C_1R_L+2C_5R_5R_Lg_m+C_5R_5} \\ \text{wo: } & \sqrt{\frac{R_5g_m+2R_Lg_m+1}{2C_1C_5R_1R_5R_Lg_m+C_1C_5R_1R_5+C_1C_5R_5R_L}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{R_5g_m+2R_Lg_m+1}{2C_1C_5R_1R_5R_Lg_m+C_1C_5R_1R_5+C_1C_5R_5R_L}}(C_1R_1R_5g_m+2C_1R_1R_Lg_m+C_1R_1+C_1R_5+C_1R_L+2C_5R_5R_Lg_m+C_5R_5)}{2\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}R_Lg_m\sqrt{\frac{R_5g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{2R_Lg_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{R_5g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{2R_Lg_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}+\sqrt{C_1}\sqrt{C_5}\sqrt{R_5}R_L\sqrt{\frac{R_5g_m}{2R_1R_Lg_m+R_1+R_L}+\frac{2R_Lg_m}{2R_1R_Lg_m+R_1+R_L}+\frac{1}{2R_1R_Lg_m+R_1+R_L}}} \\ \text{K-LP: } & \frac{R_5R_Lg_m-R_L}{R_5g_m+2R_Lg_m+1} \\ \text{K-HP: } & -\frac{R_1R_L}{2R_1R_Lg_m+R_1+R_L} \\ \text{K-BP: } & \frac{C_1R_1R_5R_Lg_m-C_1R_1R_L-C_5R_5R_L}{C_1R_1R_5g_m+2C_1R_1R_Lg_m+C_1R_1+C_1R_5+C_1R_L+2C_5R_5R_Lg_m+C_5R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-R_5g_m+1}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}} \end{aligned}$$

10.5 X-INVALID-WZ-5 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{R_L g_m + s^2 (C_1 C_5 R_1 R_5 R_L g_m - C_1 C_5 R_1 R_L) + s (C_1 R_1 R_L g_m + C_5 R_5 R_L g_m - C_5 R_L)}{g_m + s^2 (C_1 C_5 R_1 R_5 g_m + 2 C_1 C_5 R_1 R_L g_m + C_1 C_5 R_1 + C_1 C_5 R_5 + C_1 C_5 R_L) + s (C_1 R_1 g_m + C_1 + C_5 R_5 g_m + 2 C_5 R_L g_m + C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1}\sqrt{C_5}R_1R_5g_m\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+2\sqrt{C_1}\sqrt{C_5}R_1R_Lg_m\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{C_1}\sqrt{C_5}R_5\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{C_1}\sqrt{C_5}R_L\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}}{C_1R_1g_m+C_1+C_5R_5g_m+2C_5R_Lg_m+C_5} \\ \text{wo: } & \sqrt{\frac{g_m}{C_1C_5R_1R_5g_m+2C_1C_5R_1R_Lg_m+C_1C_5R_1+C_1C_5R_5+C_1C_5R_L}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{C_1C_5R_1R_5g_m+2C_1C_5R_1R_Lg_m+C_1C_5R_1+C_1C_5R_5+C_1C_5R_L}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}(C_1R_1g_m+C_1+C_5R_5g_m+2C_5R_Lg_m+C_5)}{\sqrt{C_1}\sqrt{C_5}R_1R_5g_m\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+2\sqrt{C_1}\sqrt{C_5}R_1R_Lg_m\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{C_1}\sqrt{C_5}R_5\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}+\sqrt{C_1}\sqrt{C_5}R_L\sqrt{\frac{g_m}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L}}} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & \frac{R_1R_5R_Lg_m-R_1R_L}{R_1R_5g_m+2R_1R_Lg_m+R_1+R_5+R_L} \\ \text{K-BP: } & \frac{C_1R_1R_Lg_m+C_5R_5R_Lg_m-C_5R_L}{C_1R_1g_m+C_1+C_5R_5g_m+2C_5R_Lg_m+C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \sqrt{\frac{g_m}{C_1C_5R_1R_5g_m-C_1C_5R_1}} \end{aligned}$$

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